

AI-based Training Manager for AOI? It's already here.



Delvitech's Training Manager brings neural networks and AI directly into the hands of operators, enabling real-time retraining of AOI models. Faster, easier, and more accurate quality control. All without slowing production.

Train your AOI platforms faster, smarter, and easier

The adoption of AI-driven optical inspection systems facilitates adaptability

The manufacturing landscape is evolving at record speed. Electronics, automotive, medical industries, and many other manufacturing sectors need AOI systems that don't just detect spot anomalies with precision but also adapt continuously to new challenges.

This is where deep learning comes into play. Convolutional neural networks (CNNs), trained on visual datasets, detect subtle defects that traditional methods often miss. Beyond improving first-pass yield, these AI-powered systems reduce false positives, limit unnecessary interventions, and bring real-time intelligence to inspection lines. Now, personnel can continuously train and evolve these models without starting from scratch each time.

AI and AOI? From slow training to quick retraining, with better results

In the past, updating AOI systems required data scientists, long training cycles, and carefully annotated datasets. This slowed production and increased costs. Modern AI-based inspection platforms now offer intuitive tools to annotate and train models quicker. Operators can teach systems to recognize new defects or components without disrupting production or waiting for centralized programming teams.

Economic and operational benefits of intelligent inspection

Research shows that manufacturers who have adopted AI-native AOI and SPI systems report clear returns: up to 30% faster inspection cycles and a 25% reduction in product returns[1][2]. When AI takes on the heavy lifting, sorting normal from anomalous patterns, quality control teams are freed up for more strategic tasks, whether that's process improvement, analytics, or innovation.

Training Manager: Correct, organize, and train, without interrupting production

That's where Delvitech's Training Manager stands out. Training Manager is built into the Neith software, giving operators the power to annotate, correct, and retrain

models in just minutes. Through a web-based interface, they can:

- Upload and label images
- Correct misclassifications
- Organize annotated datasets
- Train and manage multiple neural networks
- Track training history for traceability

Training Manager currently supports recognition for:

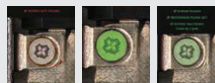
- Component bodies, SMT and also mechanical parts such as rivets and screws
- Pins
- Solder joint quality

It's a comprehensive environment where both classification and semantic segmentation are supported. Users can draw detailed polygons to define specific features, like a screw head or a solder paste deposit, then train a model to identify them with high accuracy. Even just a single image is enough to trigger a meaningful retraining cycle, letting your team correct edge cases and rare defects on the fly. Sessions run in under a minute on offline stations, keeping production lines fully operational.

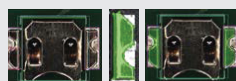
Real-world use cases

How does this play out on the production line?

Body segmentation: When the system misses a non-standard component like a screw, the operator can label it using the Annotation Tool. After training, the network not only detects the screw but also analyzes details like proper fastening.



Pin Segmentation: Initially missed, a pin is manually annotated with the Annotation tool and added to training. The model now recognizes it across the board.



AI 4 AOI

AI-Native. Future-proof by design.

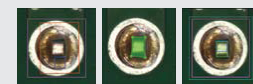
Paste deposit detection: The system initially misses the solder paste deposits, lacking prior training on them. With the Annotation tool, the operator labels just a few deposits and starts training. The network then accurately and autonomously detects all deposits.



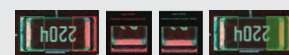
Wire bonding: Same as the paste deposit detection, with a few quick labels, the system learns to detect previously unseen wire bonds in under a minute.



Pin in Paste Segmentation: By labeling a few key examples with the Annotation tool, the operator starts the training process over a pin-in-paste scenario. Afterward, the network accurately identifies all features.



Solder joint correction: When lighting conditions lead to a misclassification of a solder joint, the operator can manually reclassify it as good, and this correction is absorbed by the neural network.



Benefits that scale

Manufacturers adopting AI-native AOI with Training Manager report tangible gains: faster inspection cycles, reduced product returns, and fewer interventions by technicians. More importantly, it empowers teams to focus on process improvement and innovation rather than repetitive error correction.

A tool designed for operators

Unlike other systems that depend on external experts, Training Manager puts control directly in the hands of your operators. No coding, no waiting, no production downtime, just smarter inspection powered by AI.

[1]Intellic (2025). Quality Inspection Trends in Manufacturing Using AI in 2025.

<https://intellic.com/quality-inspection-trends-in-manufacturing-using-ai-in-2025>

[2]Data Bridge Market Research. Global Automated Optical Inspection Market Report.

<https://www.databridgemarketresearch.com/reports/global-automated-optical-inspection-market>